

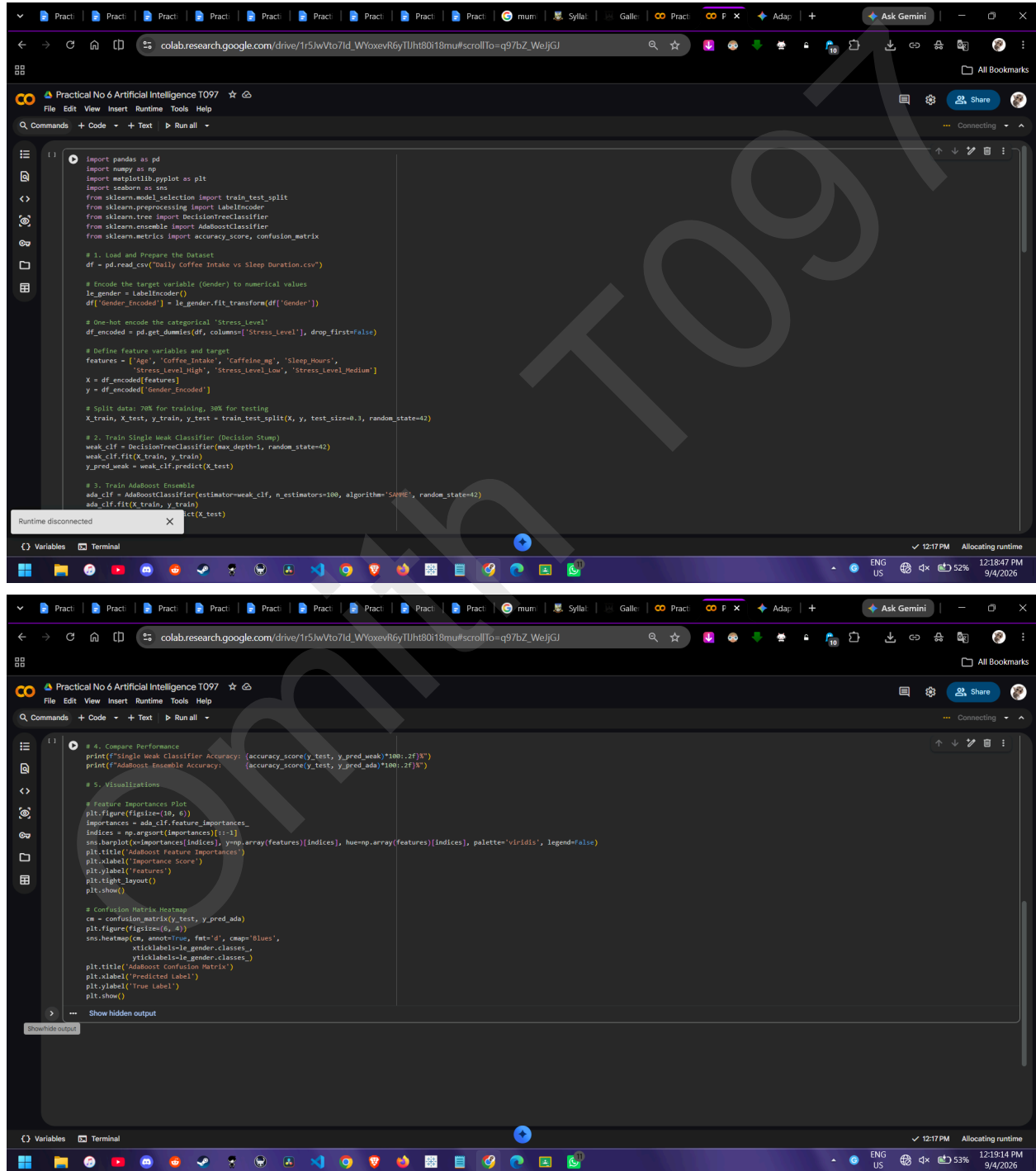
SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No 6

Aim: Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

Code:



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# 1. Load and Prepare the Dataset
df = pd.read_csv('Daily Coffee Intake vs Sleep Duration.csv')

# Encode the target variable (Gender) to numerical values
le_gender = LabelEncoder()
df['Gender_encoded'] = le_gender.fit_transform(df['Gender'])

# One-hot encode the categorical 'Stress_Level'
df_encoded = pd.get_dummies(df, columns=['Stress_Level'], drop_first=False)

# Define feature variables and target
features = ['Age', 'Coffee_Intake', 'Caffeine_mg', 'Sleep_Hours',
            'Stress_Level_High', 'Stress_Level_Low', 'Stress_Level_Medium']
X = df_encoded[features]
y = df_encoded['Gender_encoded']

# Split data: 70% for training, 30% for testing
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# 2. Train Single Weak Classifier (Decision Stump)
weak_clf = DecisionTreeClassifier(max_depth=1, random_state=42)
weak_clf.fit(X_train, y_train)
y_pred_weak = weak_clf.predict(X_test)

# 3. Train AdaBoost Ensemble
ada_clf = AdaBoostClassifier(estimator=weak_clf, n_estimators=100, algorithm='SAMME', random_state=42)
ada_clf.fit(X_train, y_train)
y_pred_ada = ada_clf.predict(X_test)

# 4. Compare Performance
print(f'Single Weak Classifier Accuracy: {accuracy_score(y_test, y_pred_weak)*100:.2f}%')
print(f'AdaBoost Ensemble Accuracy: {accuracy_score(y_test, y_pred_ada)*100:.2f}%')

# 5. Visualizations

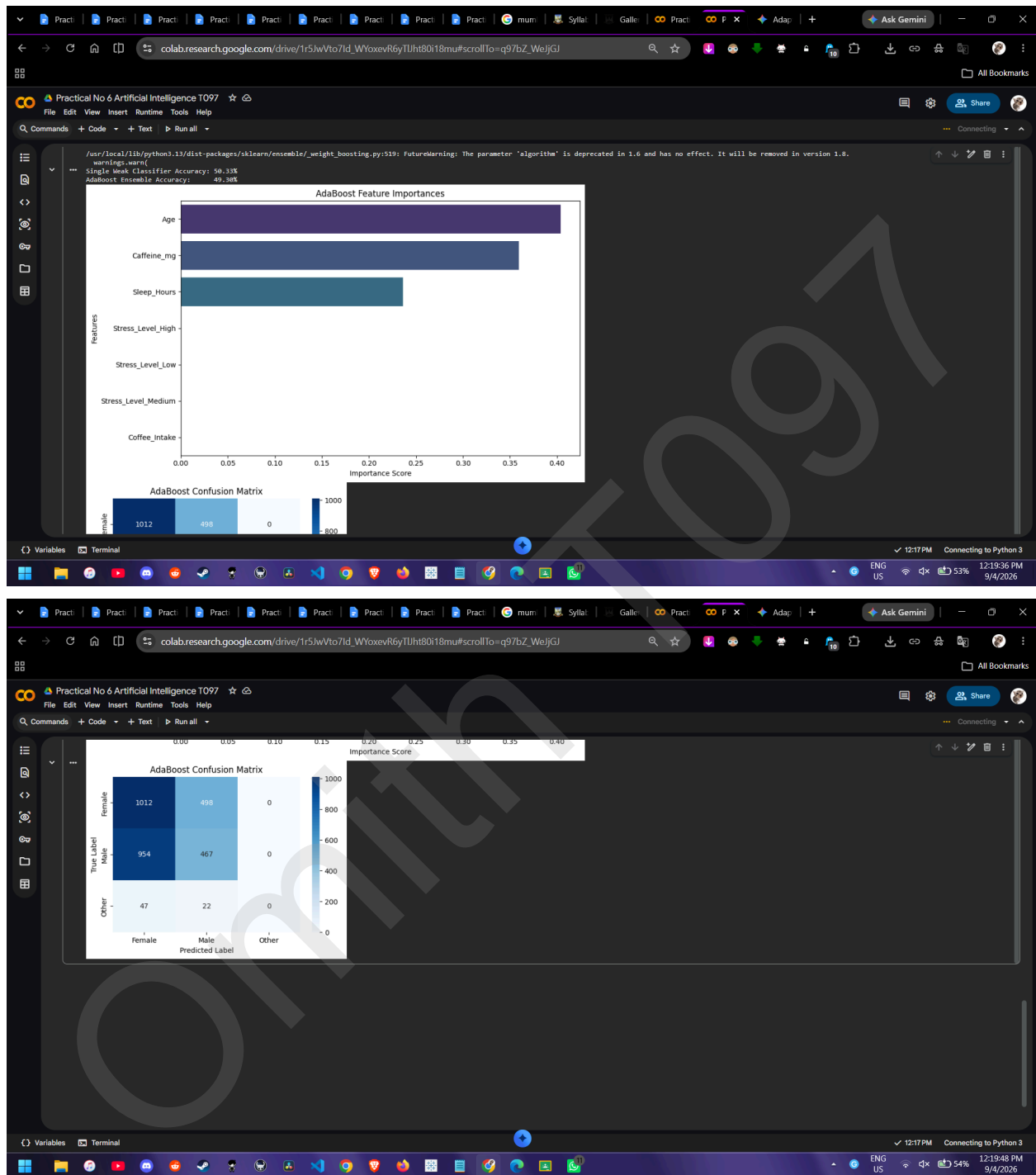
# Feature Importances Plot
plt.figure(figsize=(10, 6))
importances = ada_clf.feature_importances_
indices = np.argsort(importances)[::-1]
sns.barplot(x=importances[indices], y=np.array(features)[indices], hue=np.array(features)[indices], palette='viridis', legend=False)
plt.title('AdaBoost Feature Importances')
plt.xlabel('Importance Score')
plt.ylabel('Features')
plt.tight_layout()
plt.show()

# Confusion Matrix Heatmap
cm = confusion_matrix(y_test, y_pred_ada)
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=le_gender.classes_,
            yticklabels=le_gender.classes_)
plt.title('AdaBoost Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```

SHETH L.U.J & SIR M.V COLLEGE

Data Visualization with Power BI and Tableau

Output:



Omith Thilakan
T097